

Deep Learning (Fall 2025)

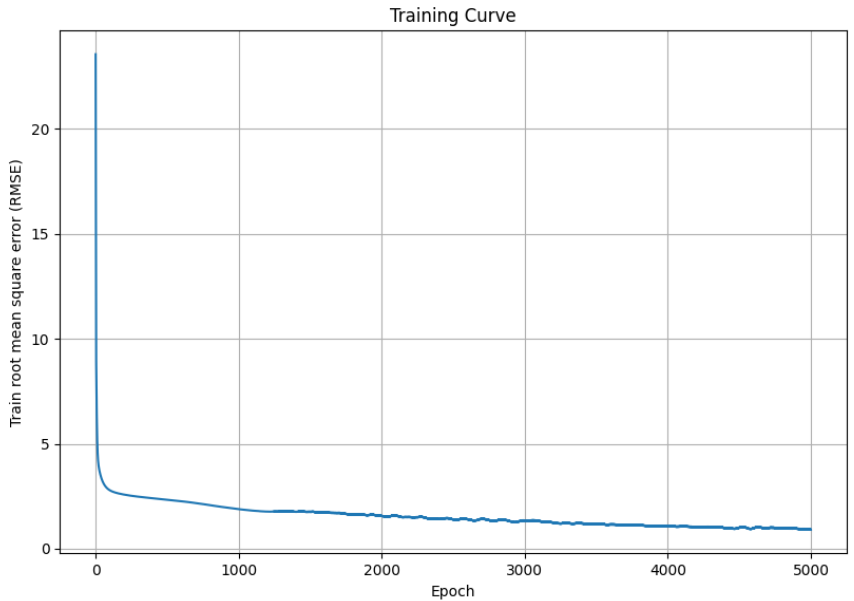
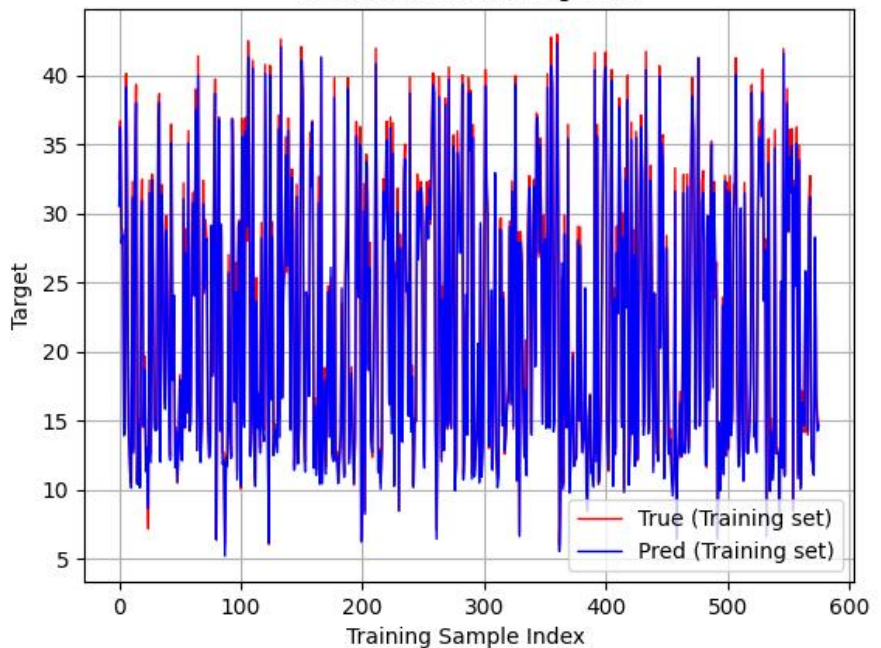
Homework 1

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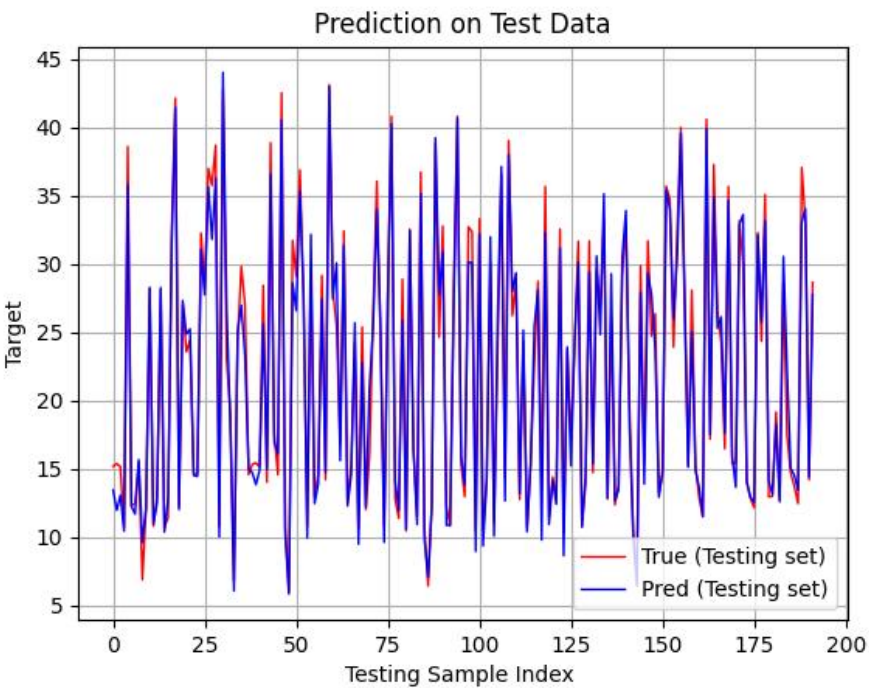
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(1) Regression

Network Architecture	16- 64- 32- 1
Learning Curve	 Training Curve The plot shows the training root mean square error (RMSE) on the y-axis (ranging from 0 to 20) against the number of epochs on the x-axis (ranging from 0 to 5000). The error starts at approximately 22 at epoch 0 and drops sharply to about 3 by epoch 100. It then gradually decreases to around 1 by epoch 5000, showing a smooth, continuous decline.
Training RMSE	0.9253
Test RMSE	1.6272
Learning Rate	0.003
Epoch	5000
Batch	Full batch
Activation Function	ReLU
Regression Result with Training Labels	 Prediction on Training Data The plot shows the target values (y-axis, ranging from 5 to 40) against the training sample index (x-axis, ranging from 0 to 600). Two lines are plotted: 'True (Training set)' (red) and 'Pred (Training set)' (blue). The lines are highly oscillatory and overlap significantly, indicating a good fit. The legend in the bottom right corner identifies the lines.

Regression Result
with Test Labels

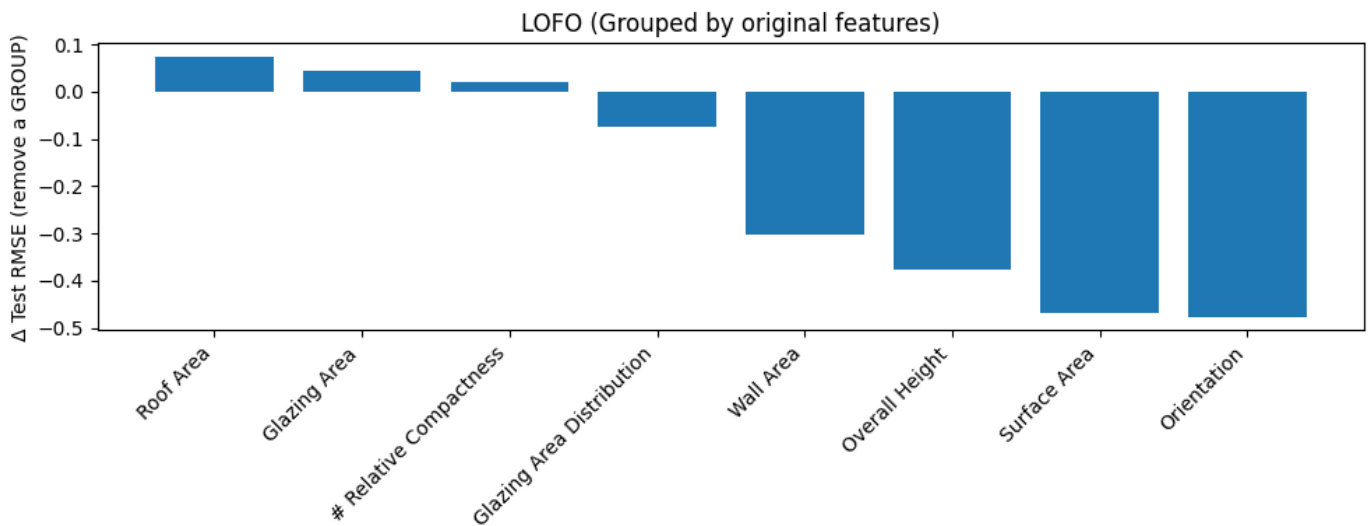


Methods of choosing the most important features:

Leave-One-Feature-Out:

Calculate the importances of a set of features based on a metric of choice, for a model of choice, by iteratively removing each feature from the set, and evaluating the performance of the model, with a validation scheme of choice, based on the chosen metric (refers to $\Delta\text{Test_RMSE}$ in this section). The formula of $\Delta\text{Test_RMSE}$ is given below. The metric tells us that if a feature is essential, the removal will increase $\Delta\text{Test_RMSE}$. On the other hand, if the feature isn't crucial, the removal may make $\Delta\text{Test_RMSE}$ decrease.

$$\Delta\text{Test_RMSE} = \text{RMSE}_{\text{remove feature}} - \text{RMSE}_{\text{all}}$$

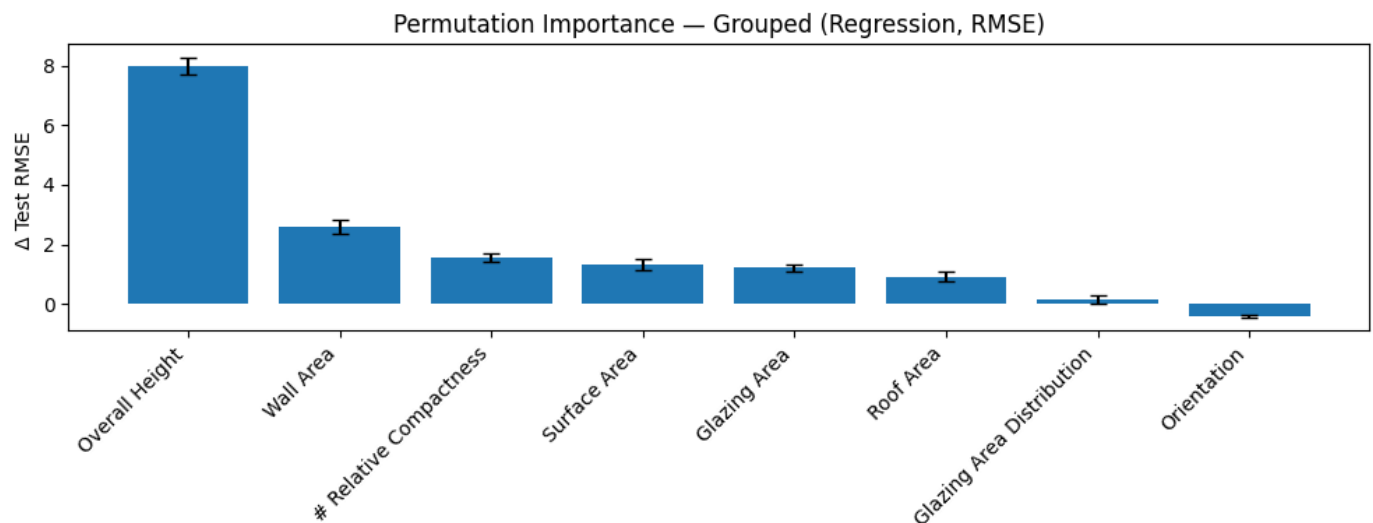


From the diagram, we can see that the removal of “*Roof Area*”, “*Glazing Area*”, and “*Relative Compactness*” are the top three causes of increment in the error rate.

Permutation Feature Importance:

Measures the increase in the prediction error (referred to as ΔRMSE in this section) of the model after permuting the feature values, which breaks the relationship between the feature and the actual outcome. The formula of the ΔRMSE is given below. The metric tells how the performance is after a feature has been permuted. The higher ΔRMSE , the more critical the feature is for the model.

$$\Delta\text{RMSE} = \text{RMSE}_{\text{permuted}} - \text{RMSE}_{\text{baseline}}$$



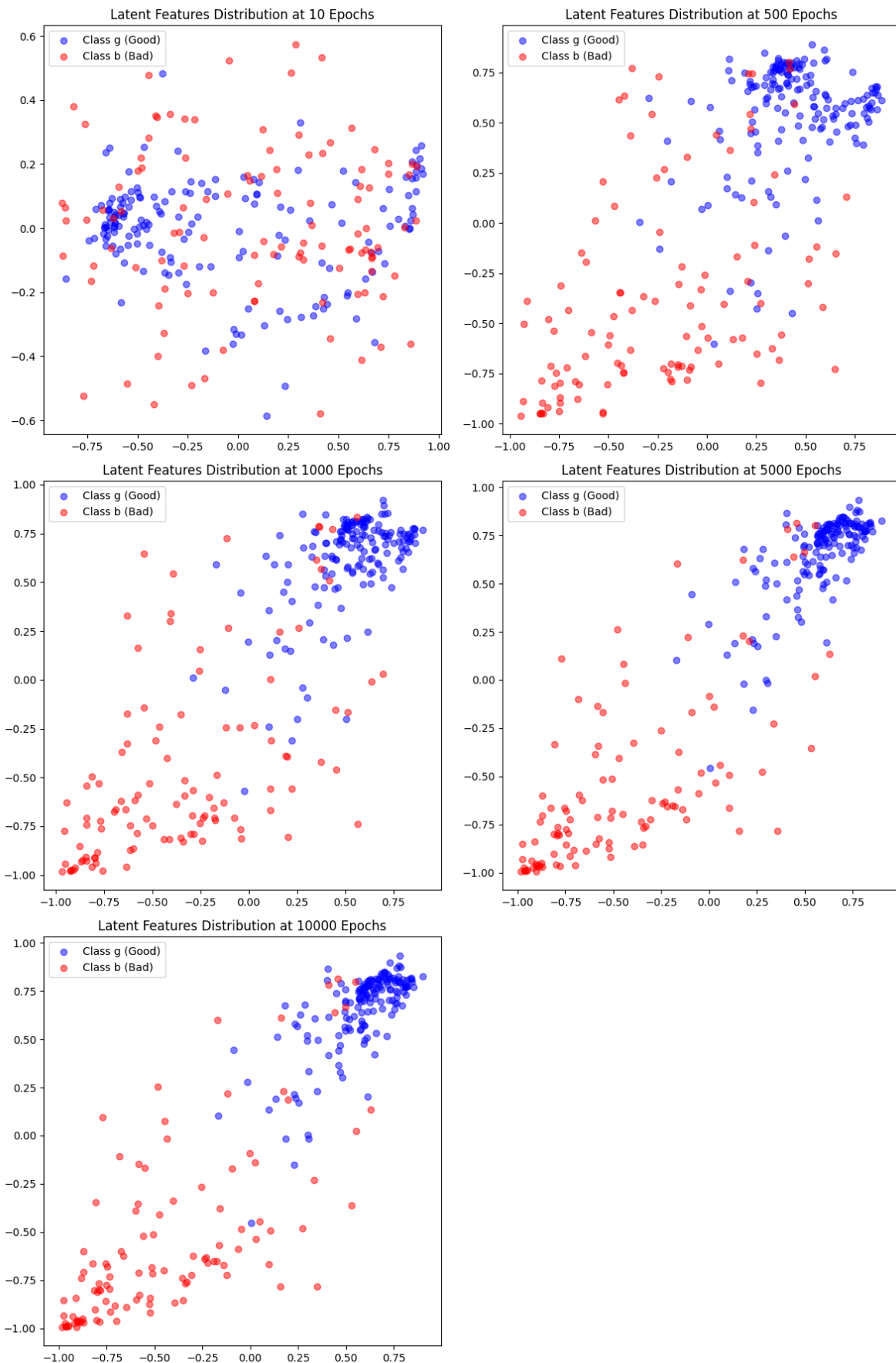
From the diagram, we can tell that the removal of “*Overall Height*”, “*Wall Area*”, and “*Relative Compactness*” are the top three causes of increment in the error rate, not to mention the fact that the error between different measurements (the black bar) exists.

(2) Classification

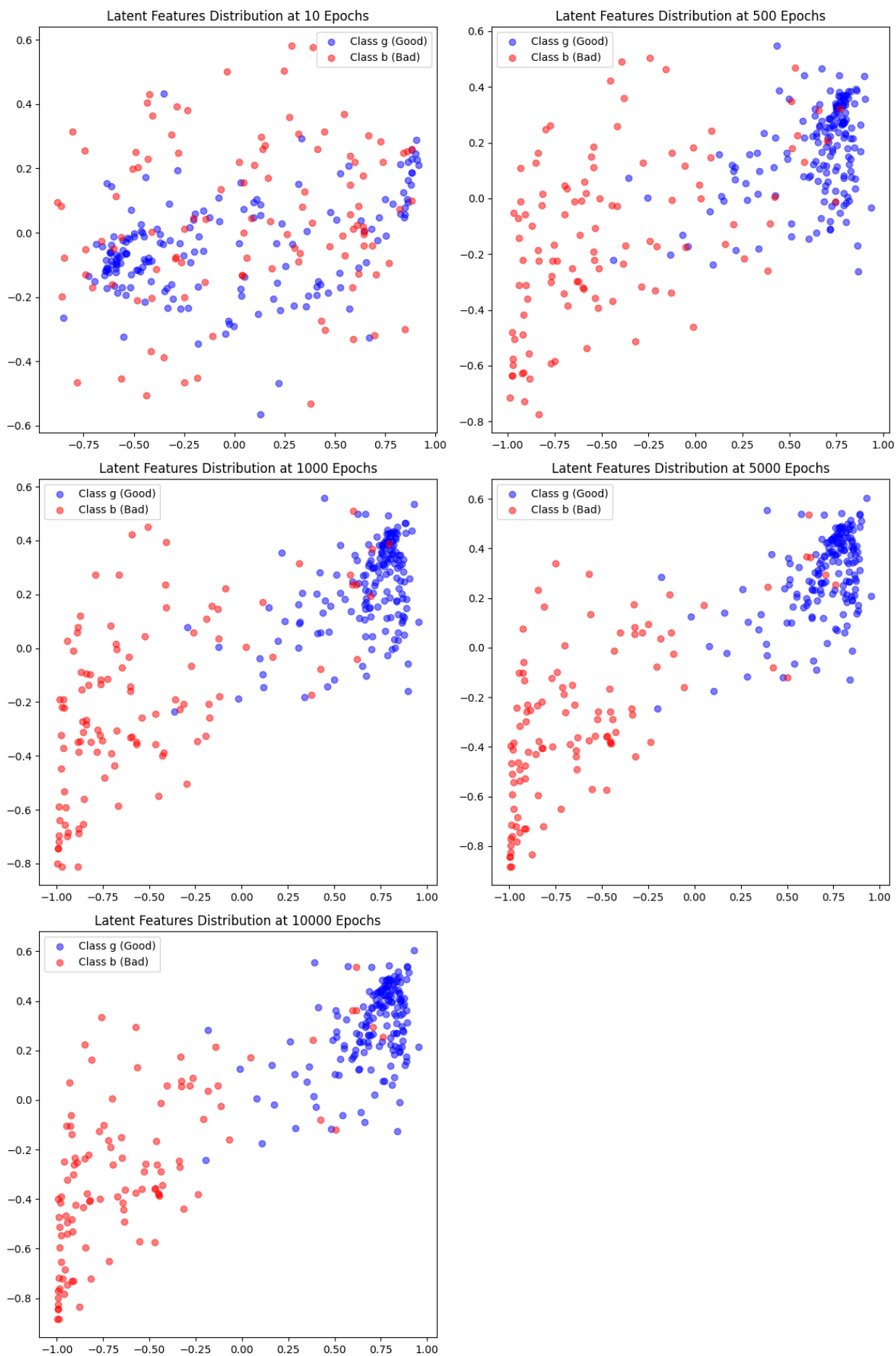
Network Architecture	34 - 128- 32- 2- 4- 1
Learning Curve	Learning Curve (34-128-32-2-4-1)
Training Error Rate	0.0500
Test Error Rate	0.1408
Learning Rate	0.004(multiply by 0.65 for each 500 epoch)
Epoch	10000
Batch	Full Batch
Activation Function	tanh (for hidden layer) sigmoid (for output layer)

Compare the results of choosing different numbers of nodes in the layer:

Network Architecture: 34- 128- 32- 2- 4- 1



Network Architecture: 34- 128- 32- 2- 8- 1



Network Architecture: 34- 128- 32- 2- 16- 1

